MIL-C-48411/6A(MU)
1 December 1975
SUPERSEDING
MIL-C-48411/6(MU)
25 April 1974

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746270
TIMING-2

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements for Timing-2.

2. APPLICATION DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746270	Circuit Card Assembly, Timing-2

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Definition.- This assembly is the second (2) of three (3) single ended printed circuit boards consisting of gates, inverters, flip-flops, and comparators necessary to provide all timing pulses to initiate the various maneuvers performed by the aircraft.

3.1.1 Major component list.- The component of this assembly shall be as specified on the Engineering Parts List (EPL) 11746270.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with a 5 VDC power supply voltage.

3.2.1.2 1st Vertical Gate - Up Clock.- With the 18 logic inputs applied to the 1st vertical gate- up clock as shown in Table I, and when outputs are specified, output Pin 22 shall be either Logic 0 or Logic 1, and output Pin 39 Logic 1 or a 60 Hz square wave clock pulse.

3.2.1.3 2nd Vertical Gate - Down Clock.- With the 18 logic inputs applied to the 2nd vertical gate - down clock as shown in Table II, and when outputs are specified, output Pins 22, 10, and 39 shall be either Logic 0 or Logic 1, output Pin 41 Logic 1 or a 60 Hz square wave clock pulse.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on Drawing 11746270 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746270 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

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4.2 Quality Conformance Inspection.- This assembly shall be inspected and tested in accordance with 4.2 or specification MIL-C-48411.

4.2.1 Test equipment.- (see FigureI)

4.2.1.1 Power supply.- The power supply shall be 10.00 ± 0.05 VDC, 100 mA or greater and a ripple of less than 10 m V RMS or 100 m V peak to peak.

4.2.1.2 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanoseconds or less, sweep period of 50 nanoseconds per centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.3 Dummy Loads.- The dummy loads shall be four (4) triple (3) input positive nand gates type SN54L10 or equivalent.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.1.5 Pulse generator.- The source shall be capable of producing a 60 Hz ± 1 Hz, 0.0 VDC to 5.0 VDC ± 0.1 VDC, rise time of 10 nanoseconds or less, square wave clock pulse.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these test shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mecury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All requirements shall be met with a power supply of +5.00 ± 0.05 VDC.

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 1st Vertical Gate - Up Clock Test.- The outputs of the 1st vertical gate - up clock, Pin 22 and Pin 39 shall function as shown in Table I when the stated inputs are applied.

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TABLE I

1ST VERTICAL GATE-UP CLOCK TEST

FUNCTION / PINS	INPUTS (LOGIC)																						OUTPUTS (LOGIC/WAVEFORMS)	
	23	38	33	36	31	20	26	18	12	16	8	4	6	4C	14	34	24	35					22	39
	1	0	0	1	0	1	0	0	1	0	0	1	0	1	0	1	1	0					0	1
	0	0	0	1	0	1	0	0	1	0	0	1	0	1	0	1	1	1					1	
	1	1	0	1	0	1	0	0	1	0	0	1	0	1	0	1	1	1					1	"1" "0" f = 60 ± 1 Hz
	1	0	0	1	0	1	0	0	1	0	0	1	0	1	0	1	1	1					1	"0" "1" f = 60 ± 1 Hz
	1	0	0	1	0	1	1	0	1	0	0	1	0	1	0	1	1	1					0	1
	1	0	0	1	0	1	0	0	1	0	0	1	0	1	0	1	1	1					0	1
	1	1	1	1	0	1	1	0	1	0	0	1	0	1	0	1	0	0					0	1
	1	0	1	0	1	1	0	0	1	0	0	1	0	1	0	0	0	0					1	"1" "0" f = 60 ± 1 Hz
	1	1	1	0	1	0	0	0	1	0	0	1	0	0	1	0	0	0					1	"0" "1" f = 60 ± 1 Hz
	1	1	1	0	1	0	1	0	1	0	0	1	0	0	1	0	0	0					0	1
	1	1	1	0	1	0	0	0	1	0	0	1	0	0	1	0	0	0					0	1
	1	1	1	0	1	1	0	0	1	0	0	1	0	0	1	0	0	0					0	1
	1	0	1	0	1	1	1	0	0	0	1	0	0	1	0	0	0	0					1	"1" "0" f = 60 ± 1 Hz
	1	0	1	0	1	1	0	0	0	0	1	0	0	1	0	0	0	0					1	"0" "1" f = 60 ± 1 Hz
	1	0	1	0	1	1	0	0	0	0	1	0	0	1	0	0	0	0					0	1
	1	0	1	0	1	1	0	0	0	0	1	0	0	1	0	0	0	0					0	1
	1	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	0					0	1
	1	1	1	0	1	0	0	0	0	0	1	0	0	0	1	0	0	0					1	"1" "0" f = 60 ± 1 Hz
	1	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	0					1	"0" "1" f = 60 ± 1 Hz
	1	1	1	0	1	0	0	0	0	0	1	0	0	0	1	0	0	0					0	1
	1	1	1	0	1	0	0	0	0	0	1	0	0	0	1	0	0	0					0	1

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TABLE II

2ND VERTICAL GATE - DOWN CLOCK TEST

FUNCTION \ PINS	INPUTS (LOGIC)																					OUTPUTS (LOGIC/WAVEFORMS)				
	23	38	33	36	31	20	26	18	12	16	8	4	6	40	14	34	24	35	22	10	39	41				
	1	1	1	0	1	0	1	1	0	0	1	0	0	1	0	0	0	0	0			1				
	1	1	1	0	1	0	1	0	0	1	0	0	0	1	0	0	0		1	1			f = 60 ± 1 Hz			
	1	1	1	0	1	0	1	1	0	0	1	0	0	1	0	0			1	1			f = 60 ± 1 Hz			
	1	1	1	0	1	0	1	1	1	0	0	1	0	0	1	0	0	0	0	0		1				
	1	1	1	0	1	0	1	1	1	0	0	1	0	0	1	0	0		0	0		1				
	1	1	1	0	1	0	1	1	1	0	0	1	0	0	1	0			0	0		1				
	1	0	0	1	0	1	0	1	1	0	0	1	0	1	1	0			0			1				
	1	1	1	0	1	0	1	1	1	0	0	1	0	0	1	0	0		1			0				
	1	0	0	1	0	1	0	1	1	0	0	1	0	1	1	0	0		0			1				
	1	0	0	1	0	1	0	1	0	1	0	1	0	1	1	0	1	1	0			1				
	1	0	0	1	0	1	0	0	1	0	0	1	0	1	1	0	1	1	0			1				
	1	0	0	1	0	1	0	0	1	0	0	1	0	1	1	0	1	1	1			0				
	1	0	0	1	0	1	0	0	1	0	0	1	0	1	1	0	1	1	1			1

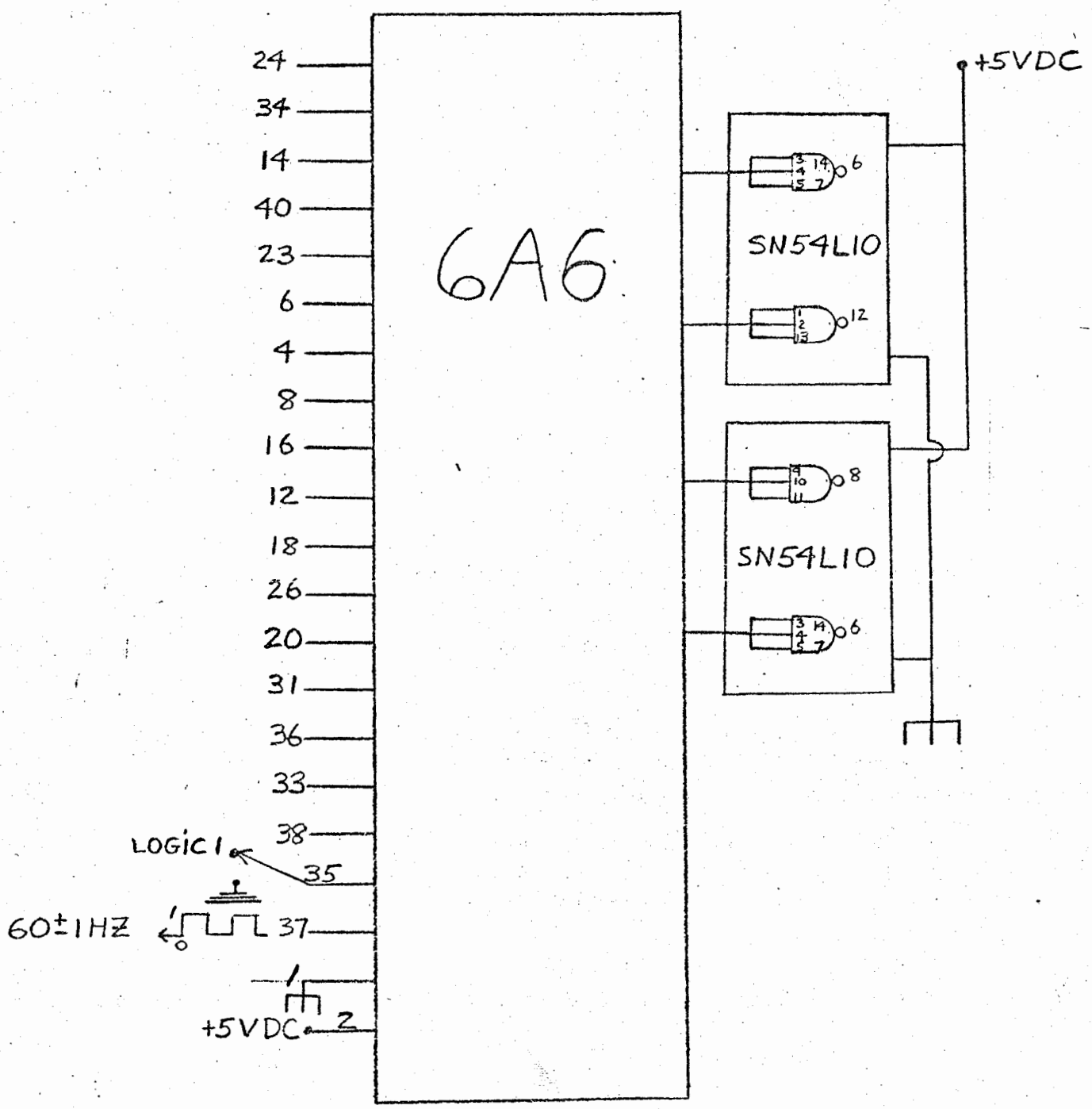


FIGURE 1

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4.2.2.3.2 2nd Vertical Gate - Down Clock Test.- The outputs of the 2nd vertical gate - down clock, Pin 22, Pin 10, Pin 39, and Pin 41 shall function as shown in Table II when the stated inputs are applied.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746270.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

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Preparing Activity:

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Project No. 1240-A601

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